

Time Series Analysis

Chapter 4: Seasonality and forecasting: SARIMA, TBATS, Prophet



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Acronyms used in this material (1/2)

- **ACF**: Autocorrelation Function
- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **AR**: AutoRegressive (model)
- **ARCH**: AutoRegressive Conditional Heteroskedasticity
- **ARIMA**: AutoRegressive Integrated Moving Average (model)
- **ARMA**: AutoRegressive Moving Average (model)
- **BIC**: Bayesian Information Criterion
- **CA**: Calendar Adjusted
- **DHR**: Dynamic Harmonic Regression
- **DM**: Diebold–Mariano (test)
- **ENTSO-E**: European Network of Transmission System Operators for Electricity
- **ETS**: Error, Trend, Seasonality (exponential smoothing state space models)
- **EU**: European Union
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GDP**: Gross Domestic Product
- **HEGY**: Hylleberg–Engle–Granger–Yoo (seasonal unit-root test)
- **HICP**: Harmonised Index of Consumer Prices
- **HLN**: Harvey–Leybourne–Newbold (small-sample correction of the DM test)

Acronyms used in this material (2/2)

- **KPSS**: Kwiatkowski–Phillips–Schmidt–Shin (test)
- **LB**: Ljung–Box (test)
- **MA**: Moving Average
- **MAE**: Mean Absolute Error
- **MAPE**: Mean Absolute Percentage Error
- **MASE**: Mean Absolute Scaled Error
- **MSTL**: Multiple Seasonal-Trend decomposition using Loess
- **NSA**: Not Seasonally Adjusted
- **OCSB**: Osborn–Chui–Smith–Birchenhall (seasonal unit-root test)
- **PACF**: Partial AutoCorrelation Function
- **RMSE**: Root Mean Squared Error
- **SARIMA**: Seasonal ARIMA (model)
- **SCA**: Seasonally and Calendar Adjusted
- **SEATS**: Signal Extraction in ARIMA Time Series
- **STL**: Seasonal and Trend decomposition using Loess
- **TBATS**: Trigonometric seasonality, Box–Cox, ARMA errors, Trend, Seasonal components (model)
- **TRAMO-SEATS**: Time series Regression with ARIMA noise, Missing values and Outliers; Signal Extraction in ARIMA Time Series
- **US**: United States
- **UTC**: Coordinated Universal Time (Temps Universel Coordonné)
- **VAT**: Value Added Tax

Today's question and route

- **Question**: many economic series repeat a pattern every year, week or day; how do we model this pattern, forecast with it, and decide which forecast is better?
  - three answers: **difference** the pattern away (SARIMA), **model** it with regressors (Fourier terms, TBATS, Prophet), or **remove** it (seasonal adjustment)
- **Route** of the chapter
  - seasonal differences and seasonal unit roots (HEGY, Canova–Hansen, OCSB)
  - SARIMA$(p, d, q)(P, D, Q)_s$: identification, estimation, diagnostics, forecasts; the airline model and Romanian GDP
  - seasonal adjustment in practice (X-13ARIMA-SEATS, Eurostat), Fourier terms and calendar effects (Orthodox Easter)
  - multiple seasonality in electricity load: MSTL, dynamic harmonic regression, TBATS, Prophet
  - forecast evaluation (time-series cross-validation, Diebold–Mariano test) and forecast combination
- It builds on Chapter 0 (components, Holt–Winters, MASE) and Chapters 2–3 (ARMA, unit roots, ARIMA): the same ideas, at the seasonal lag s
- Seminar 4 comes before this lecture: it gives the formulas; here we derive them and use them on data

Learning outcomes

- Recognise deterministic and stochastic seasonality and choose seasonal differencing with HEGY, Canova–Hansen and OCSB
- Write, identify, estimate, check and forecast a SARIMA$(p, d, q)(P, D, Q)_s$ model, and expand its polynomials
- Read seasonally adjusted and unadjusted official series correctly, and explain what X-13ARIMA-SEATS does
- Model calendar effects (working days, Orthodox Easter) and multiple seasonality with Fourier terms, MSTL, TBATS and Prophet
- Compare forecasts with time-series cross-validation, MASE and the Diebold–Mariano test, and combine them

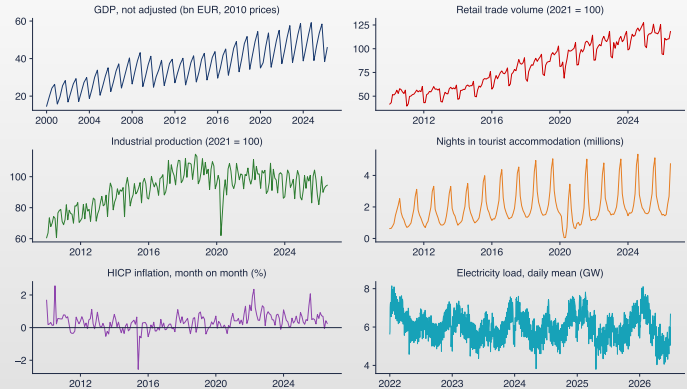
Reading and tools

- Textbook: Huang and Petukhina (2022), Ch. 4–5 (ARIMA and nonstationary models)
  - companion, free online: Hyndman and Athanasopoulos (2021), Ch. 9 (FPP3, Section 9.9), Ch. 10 (FPP3, Section 10.5), Ch. 12 (FPP3, Section 12.1) and Ch. 13
- The classic: Box, Jenkins and Reinsel (2008), Ch. 9 (seasonal models); seasonal adjustment: Ladiray and Quenneville (2001), Dagum and Bianconcini (2016), Eurostat (2024)
- Python Quantlets of this chapter: Quantlets/Ch_04
  - `statsmodels` (`SARIMAX`, `MSTL`, `ExponentialSmoothing`); optional packages `tbats` and `prophet`
- Lecture notebook: open in Google Colab
- Video course: Applied Time Series Analysis with Python

Seasonality: definition and sources

- **Seasonality**: a pattern that repeats with a fixed, known **period** s (observations per cycle)
  - $s = 4$ (quarters), $s = 12$ (months), $s = 7$ (days in a week), $s = 24$ and $s = 168$ (hours in a day and in a week)
  - unlike a **cycle**, whose length is irregular and unknown (Chapter 0)
- **Sources**
  - weather: heating in winter, the beach in August, harvests
  - the calendar: working days, public holidays, **moving holidays** such as Orthodox Easter
  - institutions and habits: the school year, tax deadlines, end-of-year bonuses, the working week
- **Additive** ($y_t = T_t + S_t + R_t$) or **multiplicative** ($y_t = T_t S_t R_t$, additive after logs): Chapter 0; here we work with logs when the swings grow with the level

Seasonality in six Romanian series



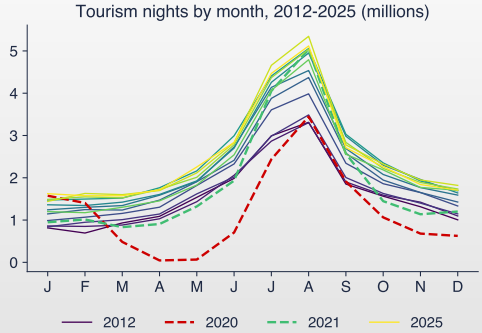
- Eurostat: GDP (quarterly, not adjusted), retail trade, industrial production, tourism nights, HICP; ENTSO-E: daily electricity load

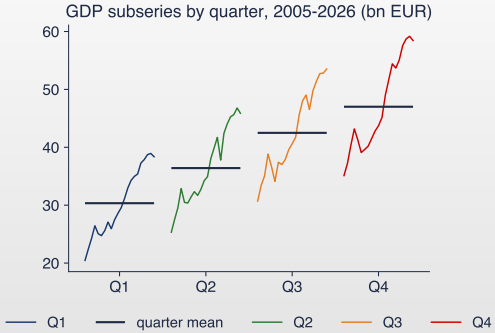
TSA_ch4_seasonal_data

Interpreting the six series

- GDP: the mean of Q4 is 57% above the mean of Q1 (2000–2026); the swing grows with the level: multiplicative, take logs
  - quarter means (bn EUR, 2010 prices): 27.8, 33.6, 39.5, 43.6: construction and agriculture in Q2–Q3
- Tourism: August has 3.7 times the nights of January (2015–2019); the 2020 lockdown breaks the pattern
- HICP: monthly inflation is lower in June (-0.37% on average, 2010–2019, fresh food) and highest in January (0.575%, new prices and excise duties), high again in October (0.49%)
- Electricity: two patterns at once, the week (dense oscillation) and the year (winter peaks): **multiple seasonality**, Section 8

Seasonal plot and subseries plot





⊡ Left: one line per year (seasonal plot); right: each quarter of GDP followed over the years, with its mean (subseries plot)

TSA_ch4_seasonal_data

Interpreting the seasonal plots

- Tourism: one shape every year, a summer peak and a winter trough
  - 2019: 5.06 million nights in August, 1.47 million in January
  - April 2020: 0.045 million (1.76 million in April 2019): an outlier, not a new season
- GDP subseries: all four quarters rise in parallel
  - a stable seasonal shape on top of a common trend
  - if the lines crossed, the seasonal pattern would be changing: stochastic seasonality (next section)



Mamaia, on the Black Sea, September 2013

Photo: Razvan Socol (2013); CC BY-SA 4.0; Wikimedia Commons

Recap: Seasonality in economic data

- Seasonality repeats with a known period s; a cycle does not
- Weather, the calendar and institutions create it; moving holidays shift it between months
- Romanian GDP, tourism, inflation and electricity load are all seasonal; electricity has several periods at once
- Seasonal and subseries plots show whether the pattern is stable

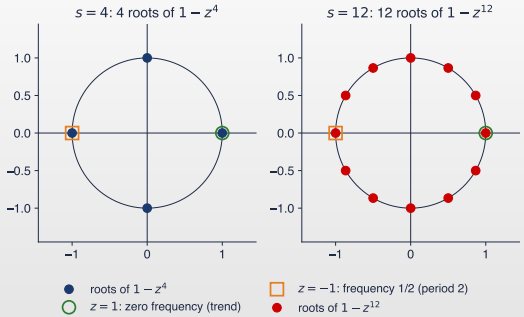
The seasonal difference

- **Seasonal difference**: $\Delta_s y_t = (1 - L^s) y_t = y_t - y_{t-s}$
  - with logs: $100\,\Delta_4 \ln Y_t$ = growth over the same quarter of the previous year, in %
- A **fixed** seasonal pattern $S_t = S_{t-s}$ plus a linear trend: $y_t = a + bt + S_t + \varepsilon_t$
  - a: the intercept; b: the slope of the trend; S_t: the seasonal effect of period t
  - $\Delta_s y_t = bs + \varepsilon_t - \varepsilon_{t-s}$: Δ_s removes the pattern and turns the trend into a constant
  - but it creates the non-invertible MA term $\varepsilon_t - \varepsilon_{t-s}$: **seasonal over-differencing** (as in Chapter 3)
- $\Delta\Delta_s = (1 - L)(1 - L^s) = 1 - L - L^s + L^{s+1}$: needed when the trend is stochastic as well
  - the order of the two differences does not matter: the operators commute

Deterministic and stochastic seasonality

- **Deterministic**: $y_t = \sum_{j=1}^{s} \gamma_j D_{jt} + u_t$, with $D_{jt} = 1$ in season j and u_t stationary
  - the seasonal means γ_j never change; the right tool: seasonal dummies or Fourier terms (Section 7)
- **Stochastic**: the **seasonal random walk** $y_t = y_{t-s} + \varepsilon_t$
  - each season is its own random walk: s separate walks interleaved
  - $\mathrm{Var}(y_t)$ grows without bound; the pattern drifts, and "summer can become winter"
  - the right tool: the seasonal difference Δ_s
- The choice matters: differencing a deterministic pattern over-differences; modelling a stochastic one with dummies leaves a unit root in the residuals

The roots of $1 - z^s$: seasonal unit roots



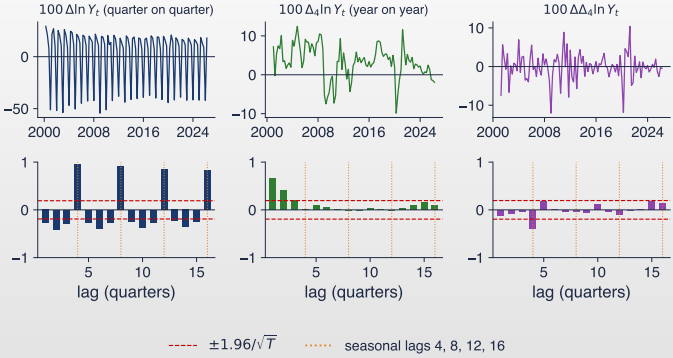
- $1 - z^4 = (1 - z)(1 + z)(1 + z^2)$: roots 1, -1, $\pm i$, all on the unit circle

TSA_ch4_seasonal_roots

Interpreting the seasonal roots

- Δ_s imposes s unit roots at once, one for each **frequency** k/s, $k = 0, \ldots, s-1$
  - quarterly: $z = 1$ (frequency 0, the trend), $z = -1$ (half a cycle per quarter: period 2 quarters), $z = \pm i$ (period 4 quarters: the annual cycle)
- A series may have some of these roots and not others: then Δ_4 over-differences at the frequencies it does not need
- Seasonal unit-root tests ask, frequency by frequency, which roots are present: the HEGY test (next slides)

Romanian GDP: regular and seasonal differences



- ln of real GDP, not adjusted, 106 quarters to 2026Q2; top: three transformations; bottom: their sample ACF up to 16 quarters

TSA_ch4_seasonal_roots

Interpreting the three transformations

- $\Delta \ln Y$: dominated by the season: ACF 0.95 at lag 4 and 0.90 at lag 8, decaying very slowly; sd 26.5%
  - the seasonal analogue of the slow ACF decay of a random walk (Chapter 3)
- $\Delta_4 \ln Y$ (year-on-year growth): sd 4.4%; ACF 0.66 at lag 1, 0.01 at lag 4: an AR-type pattern
- $\Delta\Delta_4 \ln Y$: ACF -0.11 at lag 1 and -0.40 at lag 4 (band ± 0.20)
  - one negative spike at the seasonal lag: a seasonal MA(1), the signature of the airline model (Section 3)

The HEGY test

- Hylleberg, Engle, Granger and Yoo (1990): one regression tests every seasonal root of a quarterly series
  - filters: $y_{1t} = (1 + L + L^2 + L^3)y_t$, $y_{2t} = -(1 - L + L^2 - L^3)y_t$, $y_{3t} = -(1 - L^2)y_t$
- $\Delta_4 y_t = \pi_1 y_{1,t-1} + \pi_2 y_{2,t-1} + \pi_3 y_{3,t-2} + \pi_4 y_{3,t-1} +$ deterministic terms + lags of $\Delta_4 y_t + \varepsilon_t$
  - y_{1t}, y_{2t}, y_{3t}: the series filtered so that each keeps one seasonal frequency; π_k: the coefficient of each frequency
  - H_0: $\pi_1 = 0$ (root 1), t-test; $\pi_2 = 0$ (root -1), t-test; $\pi_3 = \pi_4 = 0$ (roots $\pm i$), F-test
  - non-standard distributions, as for Dickey–Fuller: critical values from tables or by simulation
- Rejecting all three: no seasonal differencing needed; rejecting none: Δ_4 is justified



Robert Engle, a co-author of HEGY, 2017

Photo: Econterms (2017); CC BY-SA 4.0; Wikimedia Commons

Canova–Hansen and OCSB

- **Canova–Hansen** (Canova and Hansen (1995)): the null is **stable** (deterministic) seasonality
  - the seasonal analogue of KPSS (Chapter 3): rejecting means the pattern changes over time
- **OCSB** (Osborn, Chui, Smith and Birchenhall (1988)): the null is a seasonal unit root; a regression of $\Delta\Delta_s y_t$ on $\Delta_s y_{t-1}$ and Δy_{t-s}
  - the default seasonal test of `auto_arima` in the Python package `pmdarima` (`nsdiffs`)
- As in Chapter 3: tests with opposite nulls answer different questions; use them together
  - and look at the data: seasonal plots, the ACF at lags $s, 2s, 3s$, and the size of $\hat{\Theta}$ after Δ_s ($\hat{\Theta} \approx -1$ signals over-differencing)

Seasonal unit-root tests on Romanian data

HEGY, ln GDP, 2000–2019	statistic	5% critical value	decision
$t(\pi_1)$, root 1	-1.97	-3.39	not rejected
$t(\pi_2)$, root -1	-2.19	-2.77	not rejected
$F(\pi_3, \pi_4)$, roots $\pm i$	6.12	6.64	not rejected

Series (logs, 2005–2019)	D by OCSB	D by Canova–Hansen	d by KPSS
GDP (quarterly)	1	0	1
Retail trade	0	0	1
Industrial production	0	0	1
Tourism nights	0	0	1
HICP	0	0	2

- HEGY: constant, seasonal dummies and trend, 1 lag of $\Delta_4 y$ (AIC), $n = 75$; critical values from 4 000 simulations of $\Delta_4 y_t = \varepsilon_t$
- Interpretation: for GDP, HEGY and OCSB support Δ_4 while Canova–Hansen does not reject a stable pattern; with 20 years of quarters both views fit, so we use Δ_4 and let $\hat{\Theta}$ say how stable the pattern is (Section 5)
- Monthly series: both tests choose $D = 0$, a stable pattern; HICP needs $d = 2$ in logs: inflation itself is close to a unit root (Chapter 3)

Recap: Seasonal differencing and seasonal unit roots

- $\Delta_s = 1 - L^s$ has s unit roots on the circle, one per seasonal frequency
- Deterministic seasonality: dummies or Fourier terms; stochastic seasonality: Δ_s
- HEGY and OCSB test for seasonal unit roots; Canova–Hansen tests for a stable pattern
- Romanian GDP: $\Delta\Delta_4 \ln Y$ leaves one spike at lag 4, a seasonal MA

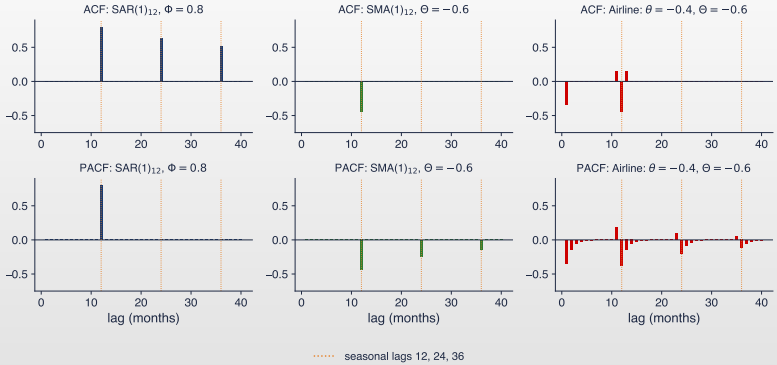
The SARIMA$(p, d, q)(P, D, Q)_s$ model

- $\phi(L)\,\Phi(L^s)\,(1-L)^d(1-L^s)^D\,y_t = c + \theta(L)\,\Theta(L^s)\,\varepsilon_t$, $\varepsilon_t \sim \text{WN}(0, \sigma^2)$
  - regular polynomials $\phi(L)$ (order p), $\theta(L)$ (order q), as in Chapter 2
  - seasonal polynomials in L^s (orders P, Q): $\Phi(L^s) = 1 - \Phi_1 L^s - \cdots - \Phi_P L^{Ps}$, $\Theta(L^s) = 1 + \Theta_1 L^s + \cdots + \Theta_Q L^{Qs}$
  - d, D: numbers of regular and seasonal differences, usually 0 or 1
- Why multiply? Box, Jenkins and Reinsel (2008): in a monthly series, January depends on December (the regular part) and on last January (the seasonal part)
  - a two-way table: months in the columns, years in the rows; one model along each direction
- Stationarity: the roots of $\phi(z)\Phi(z^s)$ outside the unit circle; invertibility: the roots of $\theta(z)\Theta(z^s)$ outside; for a seasonal AR(1) this means $|\Phi| < 1$

Worked example: the airline model

- SARIMA$(0,1,1)(0,1,1)_{12}$ with $\theta = -0.4$, $\Theta = -0.6$: $(1-L)(1-L^{12})y_t = (1-0.4L)(1-0.6L^{12})\varepsilon_t$
  - left: $y_t - y_{t-1} - y_{t-12} + y_{t-13}$; right: $\varepsilon_t - 0.4\varepsilon_{t-1} - 0.6\varepsilon_{t-12} + 0.24\varepsilon_{t-13}$
  - the coefficient of ε_{t-13}, $\theta\Theta = 0.24$, is not a new parameter: two parameters plus σ^2
- $w_t = \Delta\Delta_{12}y_t$ is an MA(13) with only four nonzero weights; its autocovariances (Chapter 2):
  - $\gamma(0) = \sigma^2(1+\theta^2)(1+\Theta^2) = 1.5776\,\sigma^2$
  - $\rho_1 = \theta/(1+\theta^2) = -0.345$; $\rho_{12} = \Theta/(1+\Theta^2) = -0.441$; $\rho_{11} = \rho_{13} = \rho_1\rho_{12} = 0.152$
  - all other autocorrelations are zero

Theoretical ACF and PACF of three seasonal models



- Monthly, $s = 12$; the airline panel is the stationary part $w_t = \Delta\Delta_{12} y_t$

TSA_ch4_sarima_theory

Interpreting the seasonal patterns

- Seasonal AR(1): the ACF decays geometrically at lags 12, 24, 36 (0.8, 0.64, 0.512); the PACF has one spike at lag 12
  - the AR rule of Chapter 2, read only at multiples of s
- Seasonal MA(1): the ACF cuts off after lag 12; the PACF decays at 12, 24, 36 (-0.242 at lag 24)
- Airline: spikes at 1 (-0.345) and 12 (-0.441), with "satellites" at 11 and 13 (0.152)
  - the satellites are the product $\rho_1\rho_{12}$: the fingerprint of the multiplicative structure

Identification of SARIMA models

After differencing	ACF	PACF
seasonal AR(P)	decays at $s, 2s, \ldots$	cuts off after lag Ps
seasonal MA(Q)	cuts off after lag Qs	decays at $s, 2s, \ldots$
regular part	lags $1, 2, \ldots$ as in Chapter 2	lags $1, 2, \ldots$ as in Chapter 2
product of both	satellites at $ks \pm j$	satellites at $ks \pm j$

- **Steps**
  - 1. Stabilise the variance (logs or Box–Cox, Box and Cox (1964)); 2. choose D (seasonal plot, HEGY, OCSB), then d (Chapter 3)
  - 3. Read P, Q at the seasonal lags and p, q at the first lags; 4. keep the orders small: $P, Q \leq 1$ almost always suffice

Estimation and model choice

- **Maximum likelihood** of the differenced series (state-space form, `statsmodels SARIMAX`)
  - $s+1$ observations are lost to $\Delta\Delta_s$; the constant is dropped when $d+D \geq 2$ (it would mean a quadratic trend)
- **Information criteria**: AICc $=$ AIC $+\frac{2k(k+1)}{n-k-1}$ and BIC, compared **only** across models with the same d and D
  - k: the number of estimated parameters; n: the number of observations after differencing
  - differencing changes the data on which the likelihood is computed
- **Automatic SARIMA**: a stepwise search over (p, q, P, Q) by AICc after choosing d and D by tests (Hyndman and Khandakar (2008))
  - in Python: `pmdarima.auto_arima(y, m=12)`; always check the residuals and compare with a benchmark
- **Diagnostics**: Ljung–Box with lags beyond s (e.g. $m = 2s$) and $m-p-q-P-Q$ degrees of freedom (Ljung and Box (1978))

Forecasting with SARIMA

- As for ARIMA (Chapter 3): write the model as an equation for y_t; replace future shocks by 0, future values by their forecasts, past shocks by residuals
  - airline: $\hat{y}_{T+h} = \hat{y}_{T+h-1} + \hat{y}_{T+h-s} - \hat{y}_{T+h-s-1} +$ (MA terms while $h \leq s+1$)
- **Eventual forecast function** of the airline model: a fixed seasonal pattern on a straight line
  - the pattern is an exponentially weighted average of past patterns, with weight $1+\Theta$ on the most recent year
  - $\Theta \to -1$: a fixed (deterministic) pattern; $\Theta = 0$: last year's pattern repeated (seasonal naive)
- Intervals widen with h (two unit roots); for $\ln y$, $\exp(\hat{y}_{T+h})$ is the forecast **median**
  - the mean is $\exp(\hat{y}_{T+h} + \sigma_h^2/2)$, slightly higher; σ_h^2: the forecast error variance at horizon h

Worked example: forecasting a seasonal MA

- Quarterly SARIMA$(0,0,0)(0,1,1)_4$: $y_t = y_{t-4} + \varepsilon_t + \Theta\varepsilon_{t-4}$, $\Theta = -0.6$
  - last year: $y_{T-3}, \ldots, y_T = 100, 120, 130, 140$; residuals $\hat{\varepsilon}_{T-3}, \ldots, \hat{\varepsilon}_T = 2, -1, 0, 3$
- $\hat{y}_{T+1} = y_{T-3} + \Theta\hat{\varepsilon}_{T-3} = 100 - 0.6 \cdot 2 = 98.8$
  - $\hat{y}_{T+2} = 120.6$, $\hat{y}_{T+3} = 130.0$, $\hat{y}_{T+4} = 138.2$
- $\hat{y}_{T+5} = \hat{y}_{T+1} = 98.8$: after one year the MA term is gone and the forecast pattern repeats
  - a large positive surprise last season ($\hat{\varepsilon}_T = 3$) lowers the forecast for that season by $0.6 \cdot 3$: partial correction

Recap: SARIMA models

- SARIMA multiplies regular and seasonal polynomials: $\phi(L)\Phi(L^s)\Delta^d\Delta_s^D y_t = \theta(L)\Theta(L^s)\varepsilon_t$
- Identification: the Chapter 2 rules at lags $s, 2s, \ldots$, plus satellites at $ks \pm j$
- Compare AICc only for the same d and D; check Ljung–Box beyond lag s
- The airline forecast: a seasonal pattern on a line, updated with weight $1 + \Theta$

Box and Jenkins and the airline passengers

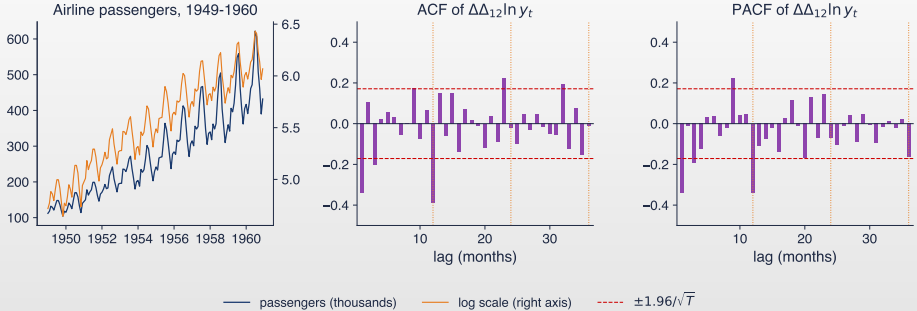
- Box, Jenkins and Reinsel (2008) (first edition 1970) modelled the monthly totals of international airline passengers, 1949–1960 (144 months)
  - growth plus a summer peak whose size grows with the level
  - their model, SARIMA$(0,1,1)(0,1,1)_{\mathbf{12}}$ for $\ln y_t$, has carried the name "airline model" ever since
- Why it lasted
  - two parameters fit many seasonal series: trade, production, retail sales, energy
  - it is the reference model of the regARIMA step of seasonal-adjustment software (Section 6)



The first Boeing 707s for Pan American, Seattle, 1958

Photo: unknown author (1958); public domain; Wikimedia Commons

Identification of the airline model



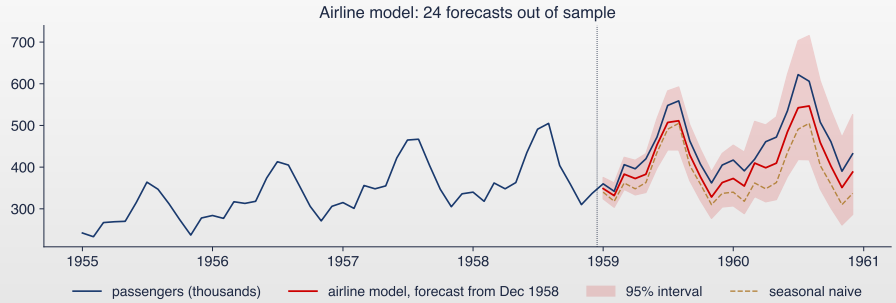
- Left: passengers and their log; middle and right: ACF and PACF of $w_t = \Delta\Delta_{12} \ln y_t$, 131 observations

TSA_ch4_airline

Interpreting the airline correlograms

- The log makes the summer swing constant: the multiplicative pattern becomes additive
- ACF of w_t: -0.34 at lag 1 and -0.39 at lag 12, outside the band ± 0.17
  - a regular MA(1) and a seasonal MA(1); lag 3 (-0.20) is borderline
  - isolated bars at lags 9, 23 and 32 also cross the band: with 36 lags, about two crossings are expected by chance
- The PACF: negative spikes at lags 1, 3 and 12, almost 0 at lag 2, and no clean cut-off after a few lags: consistent with MA terms, not AR terms
- Conclusion: SARIMA$(0,1,1)(0,1,1)_{12}$, exactly the model of Box and Jenkins

Airline model: forecasts out of sample



- Estimated on 1949–1958, forecasts for the 24 months of 1959–1960, back-transformed from $\ln y$

TSA_ch4_airline

Interpreting the airline forecasts

- 1949–1958: $\hat{\theta} = -0.342$ (0.087), $\hat{\Theta} = -0.541$ (0.105); full sample: $\hat{\theta} = -0.402$ (0.073), $\hat{\Theta} = -0.557$ (0.096)
  - full sample: residual sd 3.7% per month; Ljung–Box $Q^*(24)$ on 22 degrees of freedom, p = 0.37
- MAPE over 1959–1960: 8.5% for the airline model, 15.5% for the seasonal naive forecast (which ignores growth)
- December 1960: forecast 388, interval [286, 526], actual 432: inside, but the model under-predicts the 1959–1960 growth
  - the interval width grows from 51 (first month) to 239 (24th month): two unit roots

SARIMA models for Romanian GDP

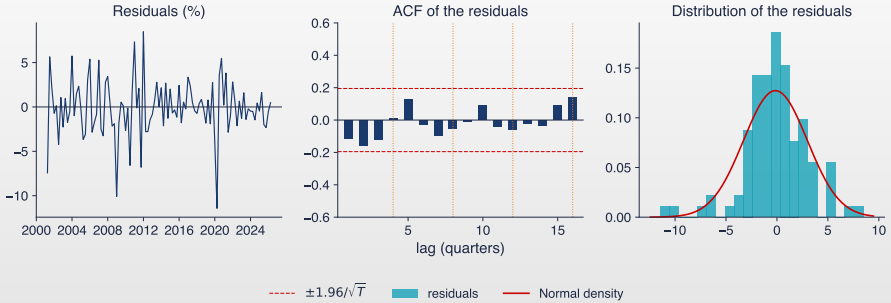
Model	AICc	BIC	LB(8), p	$\hat{\sigma}$ (%)
$(0,1,1)(0,1,1)_4$	520.1	527.6	0.157	3.05
$(0,1,0)(0,1,1)_4$	519.7	**524.8**	0.286	3.08
$(1,1,0)(0,1,1)_4$	520.7	528.3	0.156	3.06
$(1,1,1)(0,1,1)_4$	**518.3**	528.4	0.467	2.98
$(0,1,2)(0,1,1)_4$	518.8	528.8	0.575	3.00
$(0,1,1)(1,1,0)_4$	529.9	537.5	0.003	3.22
$(0,1,1)(1,1,1)_4$	522.2	532.2	0.118	3.05
$(0,1,1)(0,1,0)_4$	545.1	550.2	0.005	3.52

- $y_t = 100 \ln Y_t$, real GDP not adjusted, 2000Q1–2026Q2, $T = 106$; maximum likelihood; all with $d = D = 1$, so the criteria are comparable
- Ljung–Box on 8 lags with $8 - p - q - P - Q$ degrees of freedom

Interpreting the model table

- BIC chooses $(0,1,0)(0,1,1)_4$: only the seasonal MA term; AICc chooses $(1,1,1)(0,1,1)_4$
  - the AICc model has $\hat{\phi} = 0.78$ and $\hat{\theta} = -0.94$: almost a common factor (Chapter 2), little gain
  - the airline model: $\hat{\theta} = -0.171$ (0.109), not significant; $\hat{\Theta} = -0.585$
- A seasonal AR instead of the seasonal MA, or no seasonal term at all: much worse criteria and correlated residuals
- We keep the BIC model: one parameter, $\hat{\Theta} = -0.577$ (0.075), $\hat{\sigma} = 3.08\%$

Diagnostics of the GDP model



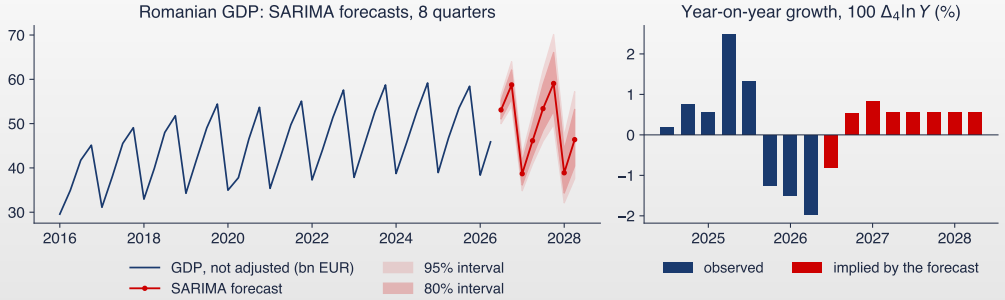
- SARIMA$(0,1,0)(0,1,1)_4$ for $100 \ln Y_t$: residuals, their ACF and their distribution

TSA_ch4_gdp_case

Interpreting the GDP diagnostics

- No autocorrelation left: $Q^*(8) = 8.55$ on 7 degrees of freedom, $p = 0.286$; $Q^*(12)$: $p = 0.526$
  - no spike at the seasonal lags 4, 8, 12, 16: the seasonal structure is captured
- Not Normal: Jarque–Bera 22.0 ($p < 0.001$); the largest residual is -11.4% in 2020Q2 (-3.6 standard deviations): the lockdown
  - without 2020, Jarque–Bera falls to 13.4: an additive-outlier dummy would be the next step

Forecasts of Romanian GDP



- SARIMA$(0,1,0)(0,1,1)_4$, 8 quarters from 2026Q3 to 2028Q2, 80% and 95% intervals; right: implied year-on-year growth

TSA_ch4_gdp_case

Interpreting the GDP forecasts

- Last value: 45.9 bn EUR in 2026Q2; forecasts 53.1, 58.7, 38.7, 46.1 bn EUR for the next four quarters
  - 95% interval for the first quarter [50.0, 56.4], for the eighth [37.6, 57.2]
- Implied year-on-year growth: -0.81%, 0.53%, 0.83%, 0.55%, then a constant 0.55%
  - the model extrapolates the recent weak growth; it knows nothing about fiscal policy or EU funds
- The seasonal pattern of the forecast is a weighted average of past years, with weight $1 + \hat{\Theta} = 0.42$ on the newest year

Recap: Airline passengers and Romanian GDP

- Logs, then $\Delta\Delta_s$, then MA terms at lags 1 and s: the airline recipe works for both series
- For GDP, BIC drops the regular MA term; AICc picks a near-redundant ARMA(1,1)
- Residuals are clean except for the 2020 outlier
- Forecasts repeat a smoothed seasonal pattern on a trend line, with widening intervals

Why official statistics adjust for seasonality

- Users want to know whether the economy **grew this quarter**, not whether Q3 is bigger than Q2
  - 2026Q1: unadjusted GDP changed by -42.1% (log points) from the previous quarter; the adjusted series by -0.06%
  - the first number is pure season; only the second says something about the business cycle
- **Seasonal adjustment**: estimate the seasonal and calendar components and remove them: $y_t^{SA} = y_t - \hat{S}_t - \hat{C}_t$ (or $y_t/(\hat{S}_t \hat{C}_t)$)
  - the classical decomposition and STL of Chapter 0 are simple versions of this
- For forecasting with SARIMA we use the **unadjusted** series: adjustment filters change the autocorrelations and are revised when new data arrive

X-11 and X-13ARIMA-SEATS

- **X-11** (US Census Bureau, 1960s; Ladiray and Quenneville (2001)): iterated moving averages
  - a centred 2×12 moving average for the trend, then 3×3 and 3×5 moving averages of each month across years for the seasonal factors
- **X-12-ARIMA** (Findley et al. (1998)) and **X-13ARIMA-SEATS** add a **regARIMA** pre-adjustment:
  - regression with SARIMA errors (often the airline model) for outliers, working days, Easter
  - ARIMA forecasts extend the series, so the symmetric filters also work at the end: smaller revisions
  - then X-11 filters or SEATS (model-based decomposition of the ARIMA model)
- EU: Eurostat and the national institutes follow Eurostat (2024), with the JDemetra+ software (X-13 and TRAMO-SEATS)



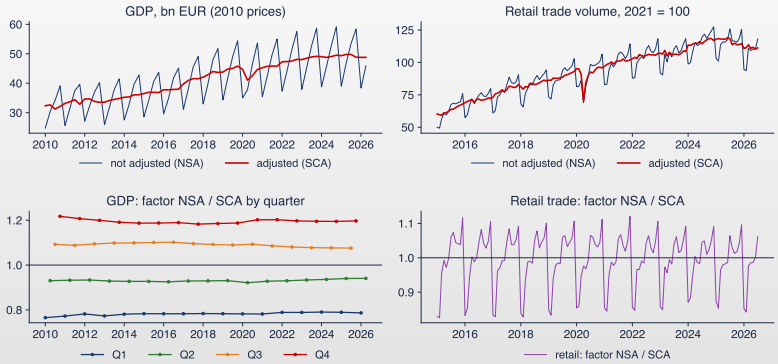
US Census Bureau, Suitland, Maryland: the home of X-11 and X-13

Photo: United States Census Bureau (2007); public domain; Wikimedia Commons

Eurostat series: NSA, CA, SA, SCA

- Each series comes in several versions (dimension `s_adj`):
  - **NSA**: not seasonally adjusted (the raw data)
  - **CA**: calendar adjusted only (working days, holidays)
  - **SA**: seasonally adjusted; **SCA**: seasonally and calendar adjusted
- Which one for which question
  - quarter-on-quarter or month-on-month growth: SCA
  - year-on-year growth: NSA or CA (the seasonal pattern cancels, the calendar does not)
  - SARIMA modelling and forecasting: NSA, with calendar regressors
- The annual totals of NSA and SCA are close (2024: 196.4 and 196.4 bn EUR): adjustment moves output between quarters, it does not create it

Unadjusted and adjusted series



- Eurostat NSA and SCA: GDP (top left) and retail trade (top right); bottom: the implied seasonal and calendar factors NSA/SCA

TSA_ch4_seasonal_data

Interpreting the adjusted series

- GDP factors 2015–2019: Q1 0.78, Q2 0.93, Q3 1.10, Q4 1.19: Q1 output is about a fifth below an average quarter
  - the factors move slowly over the years: the stochastic seasonality that $\hat{\Theta}$ measures
- Retail trade: December factor 1.10, January 0.84 (2016–2019): Christmas shopping, then the January low
- The retail factor also jumps from month to month because of working days and Easter: the calendar part of SCA (Section 7)
- The adjusted GDP shows the 2020 fall and the 2025–2026 stagnation that the raw series hides

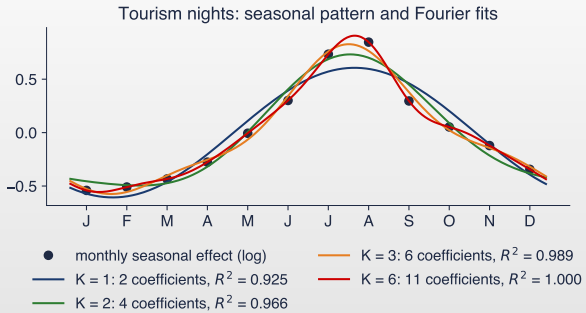
Recap: Seasonal adjustment

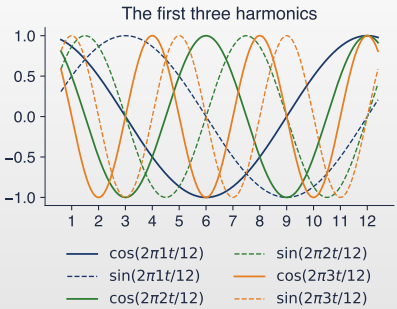
- Seasonal adjustment removes $\hat{S}_t$ and $\hat{C}_t$ to show the trend and the cycle
- X-13ARIMA-SEATS: regARIMA pre-adjustment (outliers, calendar, forecasts), then X-11 or SEATS; in the EU, JDemetra+
- Quarter-on-quarter growth: SCA; year-on-year growth: NSA or CA; SARIMA forecasting: NSA

Seasonal dummies and Fourier terms

- **Dummies**: $y_t = \beta_0 + \sum_{j=2}^{s} \gamma_j D_{jt} + u_t$: $s-1$ coefficients, any shape
  - β_0: the level of season 1; γ_j: the difference between season j and season 1
  - too many for $s = 52$ weeks, $s = 168$ hours or $s = 365$ days; impossible for $s = 365.25$
- **Fourier terms** for a period m: $\sum_{k=1}^{K} \left[\alpha_k \sin \frac{2\pi kt}{m} + \beta_k \cos \frac{2\pi kt}{m}\right]$
  - α_k, β_k: the coefficients of the k-th harmonic, a wave with k cycles per period; K: the number of sine–cosine pairs
  - $2K$ coefficients; $K = m/2$ reproduces the dummies exactly (for even m); a small K gives a smooth pattern
  - m may be non-integer, and several periods can be combined: e.g. $K = 5$ for $m = 24$ and $K = 4$ for $m = 168$ give 18 regressors instead of $23 + 167$
- K is chosen by AICc or by cross-validation; the same terms are inside TBATS and Prophet (Section 9)

Fourier approximation of a seasonal pattern





- Monthly seasonal effect of ln(tourism nights), 2012–2019, after removing a 13-month centred mean; fits with $K = 1, 2, 3, 6$; for $K = 6$, $\sin(\pi t)$ is 0 at every month and is left out (11 coefficients)

TSA_ch4_calendar_fourier

Interpreting the Fourier fits

- $K = 1$ (two coefficients) already gives $R^2 = 0.925$: the summer peak is close to a single wave
  - $K = 2$: 0.966; $K = 3$: 0.989 with six coefficients instead of eleven
- $K = 5$ (10 coefficients): $R^2 = 0.999$; $K = 6$ adds $\cos(\pi t)$, 11 coefficients, and interpolates the 12 points like 11 dummies ($R^2 = 1.000$)
- Higher harmonics add sharper detail: the August peak needs $K \geq 3$

Calendar effects and Orthodox Easter

- **Working days**: a month with more Monday–Friday non-holiday days produces and sells more
  - known in advance: a regressor that can be used in forecasts
- **Moving holidays**: Orthodox Easter falls between 4 April and 8 May
  - 5 May 2024, 20 April 2025, 12 April 2026, 2 May 2027: shopping shifts between March, April and May
  - in Python: `dateutil.easter.easter(y, EASTER_ORTHODOX)`; Pentecost is 49 days later
- **Easter regressor** (X-13 `easter[w]`): E_t = share of the w days before Easter Sunday that fall in month t
  - $w = 10$: Easter on 20 April gives $E_{\textbf{April}} = 1$; on 5 May, $E_{\textbf{April}} = 0.6$ and $E_{\textbf{May}} = 0.4$



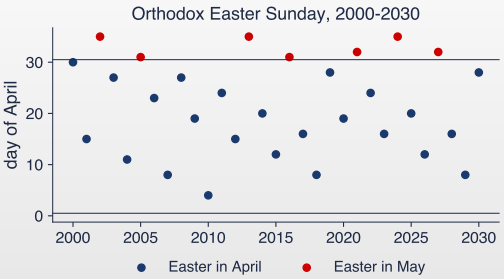
Painted Easter eggs from Romania

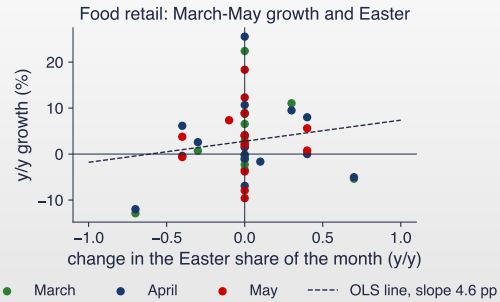
Photo: jackmac34 (Pixabay); CC0; Wikimedia Commons

Regression with SARIMA errors

- $100 \ln y_t = \beta_E E_t + \beta_W \mathrm{WD}_t + \text{outlier dummies} + u_t$, with $u_t \sim \mathrm{SARIMA}(0,1,1)(0,1,1)_{12}$
  - E_t: the Easter regressor; WD_t: the number of working days in month t; β_E, β_W: their effects, in %
  - the "regARIMA" model of X-13; estimated in one step by maximum likelihood (`SARIMAX` with exog)
  - ordinary least squares with SARIMA errors ignored would give wrong standard errors (Chapter 3, spurious regression)
- Data: Eurostat retail trade volume, Romania, not adjusted, January 2010–July 2026: food (G47_FOOD) and all retail (G47)
  - dummies for April and May 2020 (lockdown)

Easter and Romanian retail trade





- Left: Orthodox Easter Sunday as a day of April (above 30: May); right: year-on-year growth of food retail in March–May against the change in the Easter share (2020–2021 left out)

TSA_ch4_calendar_fourier

Interpreting the Easter effect

- Food retail: $\hat{\beta}_E = 5.9\%$ (se 2.3, $t = 2.6$): a month that holds the 10 days before Easter sells about 5.9% more food
  - one working day more: $+0.59\%$ (se 0.17); April 2020: -15%
- All retail: Easter 0.8% (se 1.9), not significant; working days $+1.02\%$ (se 0.15)
  - the Easter effect is in food, the working-day effect in everything
- The raw scatter is noisy (slope 4.6 percentage points): VAT changes and prices also move year-on-year growth; the regression separates the effects
- Easter fell in May in 2002, 2005, 2013, 2016, 2021, 2024: a forecaster who ignores this mistakes April and May

Recap: Fourier terms and calendar effects

- Fourier terms describe a seasonal pattern with $2K$ coefficients, for any period, even non-integer
- Working days and moving holidays are known in advance: model them as regressors
- Romanian food retail rises about 5.9% in the Easter month; total retail follows working days
- Regression with SARIMA errors: estimate regressors and dynamics together

Electricity load in Romania

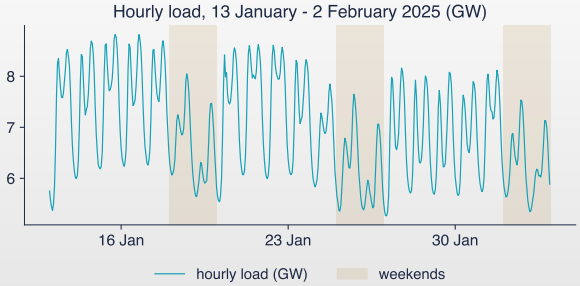
- **Load**: the electricity consumed in the system, in MW (hourly averages); it must be matched by generation at every moment
  - forecasts are needed for every hour of the next day: day-ahead markets, reserves, grid planning
- Data: ENTSO-E (the European network of transmission system operators), "monthly hourly load values", Romania
  - 39 406 hours, 2022–2026, Romanian winter time (UTC+2) to avoid the clock changes
- Load forecasting has its own competitions: GEFCom2014 (Hong et al. (2016))

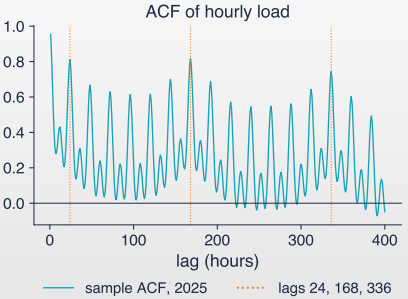


Iron Gates II hydropower plant on the Danube

Photo: Nenea hartia (2016); CC BY-SA 4.0; Wikimedia Commons

Hourly load: the day and the week





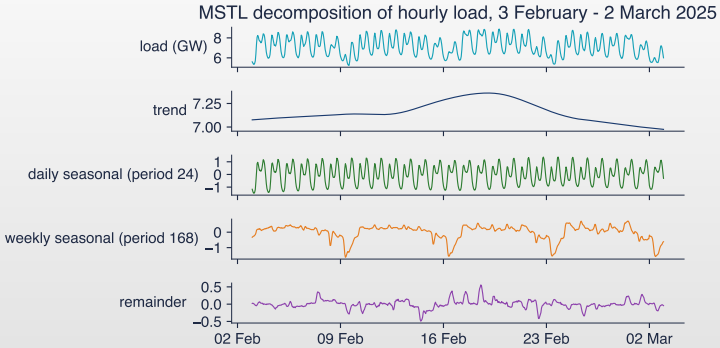
- Left: three weeks of January 2025; right: sample ACF of all hours of 2025 up to lag 400

TSA_ch4_multiple_seasonality

Interpreting the hourly load

- Daily cycle: lowest around 3:00 (5.3 GW on average in 2025), highest around 19:00 (7.3 GW)
  - ACF 0.81 at lag 24, 0.43 at lag 12
- Weekly cycle: weekends are lower; ACF 0.82 at lag 168 (one week), above the lag-24 value
  - the same hour one week ago is the best single predictor: Monday is like last Monday, not like Sunday
- And an annual cycle on top (heating and air conditioning): three periods, 24, 168 and about 8766 hours
- SARIMA is not enough: it has one integer period s, while here $s = 24$, 168 and 8766 act at once
  - a polynomial in L^{168} is slow and unstable, 365.25 is not an integer, and holidays break the weekly pattern
  - answers: MSTL, the dynamic harmonic regression (DHR), TBATS, Prophet

MSTL: one seasonal component per period



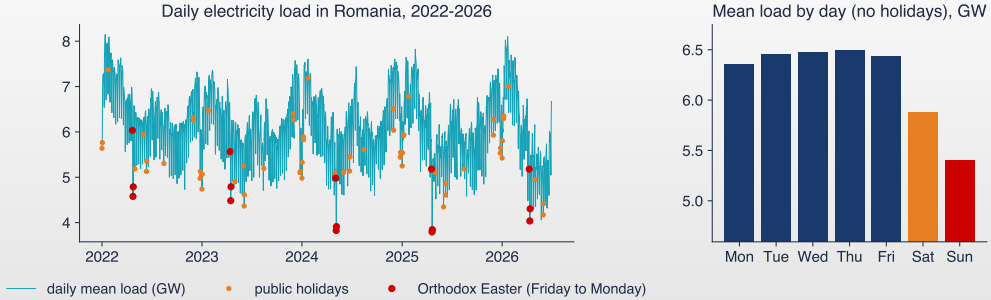
- MSTL (Bandara, Hyndman and Bergmeir (2025)) applies STL (Chapter 0) repeatedly, once for each period (24 and 168 hours), until the components settle

TSA_ch4_multiple_seasonality

Interpreting the MSTL decomposition

- Of the variance around the trend: 70% daily component, 29% weekly component, 2% remainder
  - range of the daily component in the first week: 2.8 GW; of the weekly one: 2.1 GW
- The weekly component is flat Monday to Friday and drops on Saturday and more on Sunday
- The remainder is small but not white noise: cold spells and holidays; a forecasting model adds dynamics and regressors

Daily load and public holidays



- Daily mean load, 2022–2026, 1 642 days; Romanian public holidays marked; right: mean by day of the week, holidays excluded

TSA_ch4_multiple_seasonality

Interpreting the daily load

- Week: Saturday 9% and Sunday 17% below Wednesday
  - Year: January 25% above May; a smaller summer peak (air conditioning)
- Holidays: the lowest day of the sample is Easter Sunday, 20 April 2025 (3.79 GW, mean 6.18 GW)
  - the four Easter days (Friday to Monday) average 4.62 GW; Christmas to New Year and Pentecost are also low (Pentecost Sunday was the lowest day of 2023)
- The level falls over 2022–2026; possible causes: the price shock of 2022 and rooftop solar panels that cover part of the demand behind the meter

A dynamic harmonic regression for daily load

- $y_t = \sum_{k=1}^{4} \left[\alpha_k \sin \frac{2\pi kt}{365.25} + \beta_k \cos \frac{2\pi kt}{365.25}\right] + \sum_j \delta_j D_{jt}^{\text{day}} + \lambda_1 H_t + \lambda_2 E_t + \lambda_3 X_t + u_t$
  - α_k, β_k: annual Fourier coefficients; δ_j: effect of weekday j against Sunday; $\lambda_1, \lambda_2, \lambda_3$: effects of the special days, in GW
  - annual Fourier terms; weekday dummies (Sunday is the base); H public holiday, E Easter (Friday to Monday), X 24 December – 2 January
  - $u_t \sim$ ARIMA(1,1,1), chosen by AIC: **dynamic harmonic regression** (DHR, FPP3, Section 10.5)
- Estimates, data to the first forecast origin (30 June 2025), in GW:
  - Wednesday +1.08, Monday +0.88, Saturday +0.50 relative to Sunday
  - public holiday −0.38 (se 0.02), Easter −0.58 more, Christmas–New Year −0.61
  - errors: AR 0.80, MA −0.98, $\hat{\sigma} = 186$ MW; Ljung–Box(14) p < 0.001: some dynamics left

Recap: Multiple seasonality

- Hourly load has daily, weekly and annual cycles; daily load has weekly and annual cycles
- SARIMA handles one integer period; MSTL decomposes several
- Holidays, Easter most of all, are the largest deviations: they need regressors
- DHR = Fourier terms + calendar dummies + ARIMA errors: flexible and transparent

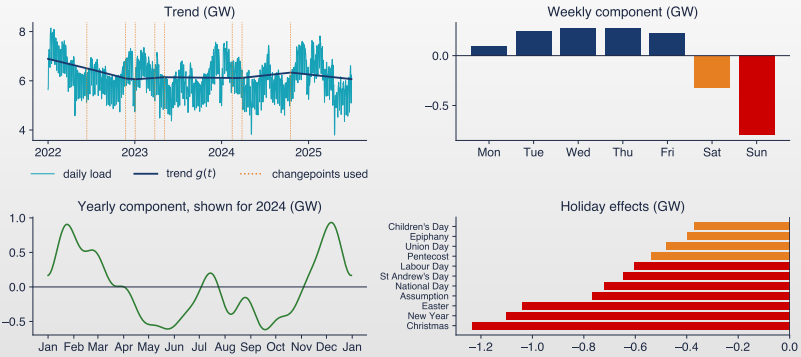
TBATS

- De Livera, Hyndman and Snyder (2011): exponential smoothing (ETS, Chapter 0) for complex seasonality; the name lists its parts
  - trigonometric seasonality (T), Box–Cox transformation (B), ARMA errors (A), trend (T), seasonal components (S)
- $y_t^{(\omega)} = \ell_{t-\mathbf{1}} + \phi b_{t-\mathbf{1}} + \sum_i s_{t-\mathbf{1}}^{(i)} + d_t$, d_t ARMA(p, q); level and trend updated as in Holt's method
  - $y_t^{(\omega)}$: the series after a Box–Cox transformation with parameter ω; ℓ_t: level; b_t: slope; ϕ: damping factor; $s_t^{(i)}$: seasonal component i
  - each period m_i has K_i harmonics $s_{j,t}^{(i)}$, each a rotating pair (cosine, sine) updated by the errors: a Fourier pattern that can evolve
- Automatic choices by AIC: Box–Cox or not, trend or damped trend or none, the K_i, the ARMA orders
  - daily load to June 2025, periods 7 and 365.25 (no Box–Cox, no trend): $K = 3$ weekly and $K = 9$ annual harmonics, ARMA(1,2), 28 states
- Limits: no regressors, hence no holidays or Easter; slow on long series; Python package `tbats`

Prophet

- Taylor and Letham (2018) (Meta): a regression in time, $y(t) = g(t) + s(t) + h(t) + \varepsilon_t$
  - $g(t)$: piecewise-linear (or logistic) trend whose slope may change at **changepoints**; a Laplace prior shrinks most changes to zero
  - $s(t)$: Fourier terms, by default $K = 10$ for the year and $K = 3$ for the week
  - $h(t)$: one effect per holiday (with optional windows around it), from a list of dates
- Estimated by Stan (maximum a posteriori); intervals simulate future changepoints and the noise
  - no autoregressive part: today's surprise does not move tomorrow's forecast
- Built for many business series with strong calendar effects and missing data; Python package `prophet`

Prophet components for daily load



⊡ Prophet with Romanian public holidays, daily load 2022 – June 2025

TSA_ch4_tbats_prophet

Interpreting the Prophet components

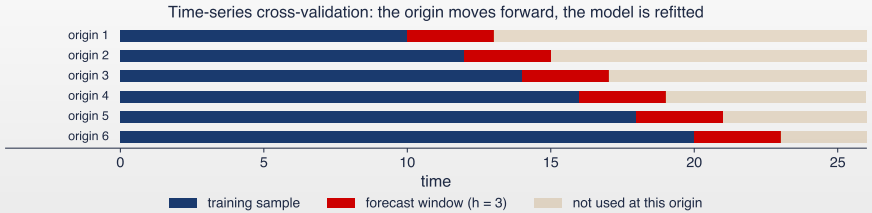
- Trend from 6.89 to 6.07 GW, flatter after 2023; 25 candidate changepoints, most shrunk to zero
  - weekly: Wednesday $+0.27$, Saturday -0.32, Sunday -0.80 GW
- Yearly: highest in early December ($+0.93$ GW, 7 December 2024), lowest in mid-September (-0.62 GW, 18 September 2024), shown for 2024
- Holidays: Christmas -1.24, New Year -1.10, Easter -1.04 GW; Children's Day only -0.37 GW
- Each component is readable and can be checked against what we know about the economy

Recap: DHR, TBATS and Prophet side by side

	DHR	TBATS	Prophet
Seasonality	Fourier, fixed	Fourier, evolving	Fourier, fixed
Several periods	yes	yes	yes
Holidays, regressors	yes	no	yes
Short-run dynamics	ARMA errors	ARMA errors	none
Trend	through the errors (ARIMA)	local (Holt)	piecewise linear
Speed	fast	slow	fast

- All three are regressions on sines and cosines at heart; they differ in what else they model
- Which forecasts best is an empirical question: the next section answers it out of sample

Time-series cross-validation



- Rolling forecast origin (Tashman (2000); FPP3, Section 5.10): fit on the data up to the origin, forecast h steps, move the origin, refit
- Expanding window (shown) or sliding window of fixed length; never use future data to fit

TSA_ch4_evaluation

Interpreting the design of our experiment

- Daily load: 26 origins every 14 days, from 30 June 2025 to 15 June 2026; horizon $h = 14$ days
  - one year of out-of-sample forecasts, every season and every holiday, Easter 2026 included
- Seven methods, all refitted at each origin
  - seasonal naive (period 7); ETS (Holt–Winters, damped trend, period 7); SARIMA$(2,0,1)(0,1,1)_7$; DHR; TBATS; Prophet
  - Combination: the simple average of the five model forecasts (all except the seasonal naive)
- The orders of SARIMA and of the DHR errors are chosen once, by AIC, on the data before the first origin: no peeking at the test period
- Cross-validation that ignores time order (random folds) is valid only for pure autoregressions with white-noise errors (Bergmeir, Hyndman and Koo (2018))

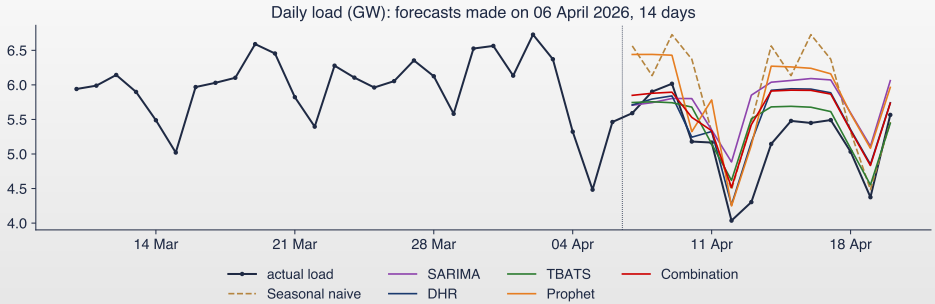
The Diebold–Mariano test

- Diebold and Mariano (1995): is the difference in accuracy between two forecasts larger than chance?
  - e_{1t}, e_{2t}: the errors of the two forecasts; L: the loss function
  - loss differential $d_t = L(e_{1t}) - L(e_{2t})$, $L(e) = e^2$ or $|e|$; H_0: $E[d_t] = 0$
- $\mathrm{DM} = \dfrac{\bar{d}}{\sqrt{\hat{V}/n}}$, $\hat{V} = \hat{\gamma}_d(0) + 2\sum_{k=1}^{h-1} \hat{\gamma}_d(k)$
  - $\bar{d}$: the mean of the n differentials; $\hat{\gamma}_d(k)$: their autocovariance at lag k; a negative DM favours forecast 1
  - h-step errors overlap, so d_t is autocorrelated up to lag $h-1$
  - asymptotically $N(0,1)$; small samples: $\mathrm{HLN} = \sqrt{\frac{n+1-2h+h(h-1)/n}{n}}$ DM, compared with $t(n-1)$ (Harvey, Leybourne and Newbold (1997))
- Our design: one value per origin, d_j = average loss difference over the 14-day window; the windows do not overlap, so $h = 1$ in the formula
  - the test compares forecasts, not models: re-estimation noise is part of what is tested

Worked example: a Diebold–Mariano test

- Five windows; window-average differences in squared error, method 1 minus method 2: $-0.30, 0.10, -0.50, -0.20, -0.40$
  - $\bar{d} = -0.26$: method 1 has the smaller losses in four windows out of five
- $\hat{\gamma}_d(0) = \frac{1}{5}\sum(d_j - \bar{d})^2 = 0.042$
  - $\text{DM} = -0.26/\sqrt{0.042/5} = -2.82$; $\text{HLN} = \sqrt{4/5} \cdot \text{DM} = -2.53$
- $|-2.53| < t_{0.975}(4) = 2.78$ ($\text{p} = 0.065$): equal accuracy is not rejected at 5%
  - the normal critical value 1.96 would have rejected: with few windows, the correction matters

Forecasts from one origin, across Easter 2026



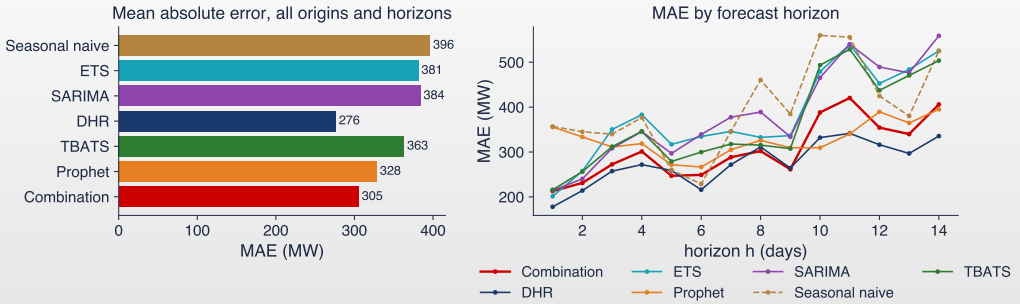
- Origin 6 April 2026, 14 days ahead; Orthodox Easter Sunday was 12 April 2026; ETS is not drawn (its MAE is on the next slide)

TSA_ch4_evaluation

Interpreting one forecast window

- MAE in this window (MW): ETS 299, TBATS 309, DHR 344, Combination 394, SARIMA 584, Prophet 616, seasonal naive 686
  - the seasonal naive method copies the previous normal week into Easter week
- Every model sees the Sunday dip; DHR and Prophet also lower Good Friday and Easter Monday, which the weekly models treat as working days
- After Easter the load stays below every forecast for the rest of the week; ETS and TBATS, whose level follows the latest days, lose least here
- One window is anecdote; the next slides average over all 26 windows

Cross-validation results for daily load



- 26 origins × 14 horizons for each method; left: overall MAE; right: MAE by horizon

TSA_ch4_evaluation

Interpreting the cross-validation results

Method	MAE	RMSE	MASE	DM against s. naive	DM against DHR
Seasonal naive	396	529	1.28	–	4.42 (<0.001)
ETS	381	492	1.23	-0.80 (0.431)	3.08 (0.005)
SARIMA	384	477	1.24	-1.23 (0.231)	3.29 (0.003)
DHR	**276**	**350**	**0.89**	-4.42 (<0.001)	–
TBATS	363	468	1.18	-1.50 (0.145)	3.02 (0.006)
Prophet	328	412	1.06	-2.99 (0.006)	2.76 (0.011)
Combination	305	380	0.99	-3.82 (<0.001)	1.89 (0.070)

- MAE and RMSE in MW; MASE (Hyndman and Koehler (2006)) = MAE / 309 MW, the in-sample MAE of the weekly seasonal naive method; DM: HLN statistic (p-value) on window-average squared errors, row method minus column method
- DHR wins; with the combination, it is the only method with MASE below 1; Prophet is the second single model: the two that know the holidays
- ETS, SARIMA and TBATS are not significantly better than the seasonal naive method

A second test: Romanian GDP

Method	MAE, $h = 1$	MAE, $h = 4$	without 2019–2020, $h = 1$	without 2019–2020, $h = 4$
SARIMA (airline)	1.69	2.70	1.32	1.89
ETS	1.76	2.84	1.41	2.44
Seasonal naive	3.81	3.74	3.80	3.49
Combination (SARIMA, ETS)	1.70	2.65	1.34	2.02

- $100 \ln Y$, errors in %; 51 quarterly origins, 2012Q4–2025Q2, expanding window; 43 origins without 2019–2020
- DM (HLN), SARIMA against ETS: $h = 1$: -1.46 ($p = 0.150$); $h = 4$: 0.50 ($p = 0.620$)
- SARIMA against the seasonal naive method: $h = 1$: -4.00 ($p < 0.001$); $h = 4$: -0.54 ($p = 0.591$)
- One year ahead, the pandemic errors dominate and no difference is significant; without them, SARIMA has the smallest MAE at $h = 4$

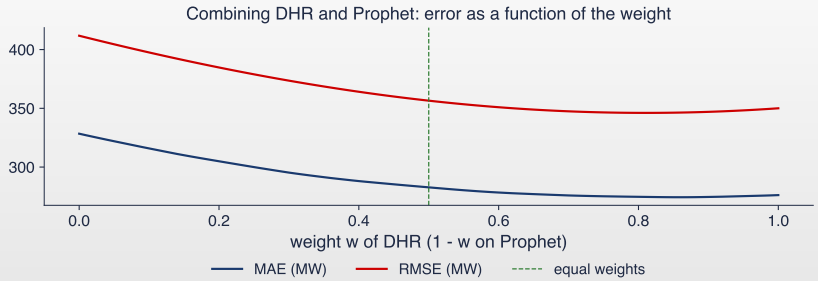
Recap: Forecast evaluation

- Evaluate out of sample with rolling origins; choose model orders before the test period
- MASE compares with the seasonal naive method and across series
- The DM test (with the HLN correction) says whether a gain is more than chance
- Daily load: the models that know the calendar (DHR, Prophet) win; for GDP, SARIMA and ETS are close

Why combining forecasts works

- Bates and Granger (1969): a weighted average of two unbiased forecasts, $f_c = wf_1 + (1-w)f_2$
  - w: the weight of forecast 1; σ_1, σ_2: the standard deviations of the two errors; ρ: their correlation
  - $\text{MSE}(w) = w^2\sigma_1^2 + (1-w)^2\sigma_2^2 + 2w(1-w)\rho\sigma_1\sigma_2$
  - best weight $w^* = \dfrac{\sigma_2^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$; the gain is large when ρ is small
- Example: $\sigma_1 = 1$, $\sigma_2 = 1.2$, $\rho = 0.75$: $\text{MSE}(0.5) = 1.060$; $w^* = 0.84$, $\text{MSE}(w^*) = 0.984$
  - highly correlated errors: the equal-weight average is worse than the better forecast ($\sigma_1^2 = 1$), and even the best weight gains little
- The **forecast combination puzzle** (Stock and Watson (2004); Smith and Wallis (2009)): estimated weights rarely beat equal weights
  - the weights must be estimated, and their sampling error eats the theoretical gain

Combining DHR and Prophet



- Cross-validation errors of the daily load; w is the weight of DHR

TSA_ch4_evaluation

Interpreting the combination

- MAE: Prophet alone (w = 0) 328 MW, DHR alone (w = 1) 276 MW, equal weights 283 MW
  - best in hindsight: $w = 0.86$, MAE 274 MW, a small gain chosen after seeing the test data
- The errors of the two methods are strongly correlated ($\rho = 0.75$): both miss the same cold spells
- The five-model average (MAE 305) is second only to DHR and better than four of its five members: cheap insurance when the winner is not known in advance
- In the M4 competition (Makridakis, Spiliotis and Assimakopoulos (2020)), most of the best methods were combinations; review: Wang et al. (2023)

Possible contribution of AI

- **Code**: a first draft of a rolling-origin loop that refits SARIMA, DHR, TBATS and Prophet and stores the errors by horizon
- **Calendars**: lists of holidays and moving feasts for other countries, to be checked against official sources
- **Explanation**: a second derivation of the ACF of the airline model, or of the eventual forecast function
- Example prompt
  - `Write Python code that forecasts daily electricity load 14 days ahead with a regression on annual Fourier terms, weekday dummies and Romanian public holidays (Orthodox Easter from dateutil) with ARMA errors, and evaluates it against SARIMA by rolling origins and a Diebold-Mariano test.`

Checks you must run

- The Easter date: Orthodox, not Western (`EASTER_ORTHODOX`); in 2024 the two differ by five weeks
- No look-ahead: features, model orders and holiday lists must be available at each origin
- The sign conventions of θ and Θ in the software, and the degrees of freedom of Ljung–Box
- Adjusted or unadjusted data: SARIMA on NSA; growth rates on the right version
- That the DM test uses losses from the same windows for both methods, and the HLN correction for small n
- Every cited reference: it must exist; check the DOI

Key takeaways

- Seasonality repeats with a known period; it can be deterministic (dummies, Fourier) or stochastic (Δ_s)
- SARIMA multiplies regular and seasonal polynomials; the airline model fits many series with two parameters
- Official data: SCA for short-run growth, NSA with calendar regressors for SARIMA forecasting
- Calendar effects (working days, Orthodox Easter, holidays) are known in advance and often matter more than the model class
- Multiple seasonality: MSTL, DHR, TBATS, Prophet; on Romanian daily load, DHR wins out of sample
- Judge forecasts by rolling origins, MASE and the DM test; combine them when the winner is uncertain

Key formulas

Quantity	Formula
Seasonal difference	$\Delta_s y_t = (1-L^s)y_t, \quad \Delta\Delta_s = 1 - L - L^s + L^{s+1}$
SARIMA	$\phi(L)\Phi(L^s)(1-L)^d(1-L^s)^D y_t = \theta(L)\Theta(L^s)\varepsilon_t$
Airline	$\rho_1 = \frac{\theta}{1+\theta^2}, \quad \rho_s = \frac{\Theta}{1+\Theta^2}, \quad \rho_{s\pm1} = \rho_1\rho_s$
Fourier	$\sum_{k=1}^{K}[\alpha_k \sin(2\pi kt/m) + \beta_k \cos(2\pi kt/m)]$
Prophet	$y(t) = g(t) + s(t) + h(t) + \varepsilon_t$
MASE	$\frac{1}{h}\sum \lvert e_{T+j}\rvert \,/\, \frac{1}{T-m}\sum \lvert y_t - y_{t-m}\rvert$
DM	$\bar{d}/\sqrt{\hat{V}/n}, \quad d_t = L(e_{1t}) - L(e_{2t})$
Combination	$w^* = \frac{\sigma_2^2 - \rho\sigma_1\sigma_2}{\sigma_1^2+\sigma_2^2-2\rho\sigma_1\sigma_2}$

Self-assessment

- **Question**: how many parameters does SARIMA$(1,1,1)(0,1,1)_{\mathbf{12}}$ have, and at which lags does ε appear?
  - **Answer**: $\phi, \theta, \Theta, \sigma^{\mathbf{2}}$, four; ε at lags 0, 1, 12, 13 (the last with coefficient $\theta\Theta$)
- **Question**: unadjusted GDP falls by 40% from Q4 to Q1. Is this a recession?
  - **Answer**: no; it is the seasonal pattern; look at the SCA series or at year-on-year growth
- **Question**: a DM test with 26 windows gives HLN $= -1.93$. Is the first forecast significantly better at 5%?
  - **Answer**: no: $|-1.93| < t_{\mathbf{0.975}}(25) = 2.06$; it is better on average, but not significantly
- **Project idea**: forecast Romanian hourly electricity load with a DHR (daily, weekly and annual Fourier terms, holidays, temperature) against TBATS, Prophet and a combination, by rolling origins and DM tests
  - Next: Chapter 5, conditional volatility: ARCH and GARCH models

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